

STATEMENT OF NEED
US Space Force
Test Set specifically for the ATCALS equipment
Current as of 20 April 2026

1) SCOPE:

The Buckley Space Force Base Radar, Airfield & Weather Systems (RAWS) unit requires exacting test sets to perform essential maintenance on Air Traffic Control and Landing Systems (ATCALS). To maintain Federal Aviation Administration (FAA) certification, these systems must undergo a flight inspection every eighteen months. During these inspections, critical parameters such as reflected radio frequency (RF) and cable phasing must be precisely measured and adjusted. Headquarters Air Force Flight Standards Agency (HQ/AFFSA) has identified the specified test requirements as uniquely suited for the ATCALS mission.

The operational readiness of ATCALS is vital for flight operations during nighttime and all-weather conditions. These systems provide aircraft with precision horizontal and vertical guidance starting from 18 nautical miles, enabling approaches down to 250 feet above the runway threshold. The proper function of these systems is critical to flight safety. The test set will also support the maintenance of ATC Communications systems.

2) REQUIREMENT:

Item Description	Quantity
Test System with ILS, DOD ATG Kit	3
Calibration Certificate with Test Data	3
5-Year Total Hardware Warranty & Calibration	3

a) Must be compatible to perform most maintenance functions for the following systems:

- i) AN/FRN-45C, Tactical Air Navigation (TACAN)
- ii) CM300 UHF & VHF Transmitter and Receiver Radios
- iii) CM350 Transceiver Radios
- iv) AN/FSC-127, Enhance Terminal Voice Switch (ETVS)
- v) AN/GRN-29/30/31, Instrument Landing System (ILS)
- vi) PRC-113, Transceiver Radio Set
- vii) Transmission Impairment Measurement Set (TIMS)
- viii) Plain Old Telephone Service (POTS)

b) Description:

- i) Must be developed for the USAF Test System in conjunction with the Air Force Flight Standards Agency to test the General Dynamics CM300/350 Receiver and Transmitter.
- ii) The test system includes everything needed to test and align the CM300/350 radios and includes required options:
 - (1) Preloaded procedure documents for UHF, VHF Low Power and VHF High Power radios
 - (2) Predefined setup files that are specifically designed for each test required
 - (3) Complete cable and adapter kit
- iii) The procedure documents have been developed directly from the Air Force

- iv) Special Maintenance Instructions for Air Traffic Control Communications Systems - AFMSI 300-1 (1 Sep 22) and the FAA Test Instructions TI 6600.32 and TI 6600.33.
- v) The procedures cover the complete set of tests required on the radios and include information on operating both the test set and the CM300/350 radios.
- vi) Designed to improve both speed of test and accuracy in the testing process, the documents are easy to navigate and understand.
- vii) Preinstalled Procedure Documents Reside on the test set without the need for a separate computer.

3) SPECIFICATIONS:

a) FULLY INTEGRATED TEST EQUIPMENT

- i) Test set that includes:
 - (1) power meter
 - (2) RF frequency counter
 - (3) Modulation analyzer
 - (4) Audio generator
 - (5) Level meters
 - (6) Spectrum analyzer
 - (7) Oscilloscope
 - (8) All in one instrument.
 - (9) (This eliminates the need to send multiple instruments to PMEL for annual calibration.)
- ii) Must be setup in the USAF PMEL system for annual calibration.
- iii) Additional specifications:
 - (1) 6 GHz Range Extension.
 - (2) Cable and Antenna Analysis (DTF / VSWR / IL / RL).
 - (3) Vector Network Analyzer functions for Insertion Loss (S21) and Group Delay.
 - (4) S-Parameter Option Upgrade. Adds S11 and S21 parameters.
 - (5) Checklist Configurator.
 - (6) Bird External Power Meter Option to Support (Does not include 5017D)

b) Accessories must include:

- i) Carrying Case, Hard Transit
- ii) Carrying Case, Soft
- iii) Internal Battery, 98 Watt-Hour, Lithium-Ion
- iv) Antenna Kit, 1 GHz
- v) Carrying Case, Backpack
- vi) Power Adapter, Automotive DC/DC (Cigarette Lighter)
- vii) External Battery Charger
- viii) GNSS/GPS SMA Mount Antenna
- ix) Microphone
- x) Accessory, Short, Open, Load
- xi) Calibration Certificate with Test Data (ISO 9001)
- xii) 3 Year; extended warranty & calibration
- xiii) 5 Year; extended warranty & calibration
- xiv) 3 Year Care Includes extended warranty, calibration, loaner and accessory replacement.

- xv) 5 Year; extended warranty, calibration, loaner and accessory replacement.
- c) USAF – Kit Cable Kit, Test and Alignment Procedures and Predefined Set Up Files
- VHF Low Power, VHF High Power and UHF Test Procedures.
 - Qty 1 – BNC-M to N-M Cable 10' (The Orange Cable)
 - Qty 2 – N-M to N-M Cables 5' (The New Blue Cables)
 - Qty 2 – N-M to BNC-M Adaptor
 - Qty 2 – Banana to BNC-M Adaptors
 - Qty 2 – BNC-M to 2 Bantam TT Jacks 0.173" 5' (The New Red Cable)
 - Qty 1 – BNC-M to BNC-M Cable 5' (The Yellow Cable)
 - Qty 2 – N-F to N-F Barrel Adaptors
 - Qty 2 – BNC-F to BNC-F Barrel Adaptor
 - Qty 2 – SMA-M to N-F
 - Qty 2 – SMA Bullet Adapter
 - Qty 2 – TNC (F) to N (F) Adaptor
 - Qty 2 – TNC (M) to N (F) Adaptor
 - Qty 2 – TNC (F) to BNC (F) Adaptor
 - Qty 2 – TNC (M) to BNC (F) Adaptor
 - Qty 2 – SMA (F) to N (F) Adaptor.

4) DELIVERY:

The contractor shall deliver the required test sets no more than 60 days after receipt of order (ARO). The required delivery date is no later than 30 July 2026. The specific delivery address will be provided upon contract award.

Upon delivery, the items will be inspected and accepted by designated government personnel.